or very low, low, average, and high twist. Yarn twist impacts the yarn's appearance, fineness, strength, and absorption.

Directions Of Twist:

Fibres can be twisted together in the clockwise or counter clockwise direction to form yarns. Yarns are twisted in the clockwise direction for "S" twist, and counter clockwise for "Z" twist. The "Z" twist is employed in a majority of the spun yarns used for fabric construction.



Directions Of Twist

4.4.2 Classification Of Yarns

Classification of yarns based on fibre length is used to broadly divide yarns into: – **Spun yarns (made from short, staple fibres) – Filament yarns (made from continuous filament fibres)**

Spun yarns are composed of short staple fibres, or long filament fibres that have been cut into short staple fibres. Spun yarns may contain fibres of the same type or a blend of different fibres. The characteristics of spun yarn depend on the amount of twist given to the fibres during spinning. A fairly high degree of twist produces strong yarn, a low twist produces softer, more lustrous yarn and a very tight twist produces crepe yarn.

Filament yarns are extruded yarns that are made of natural extruded fibres or manufactured fibres extruded through the spinneret. These yarns fibres can be broadly divided into:

1. Monofilament yarn

2. Multifilament yarn

Monofilament yarn is made from a single, relatively thicker filament fibre. Transparent sewing thread, metallic yarns, bare elastic, and fishing lines are examples of monofilament yarns.

Multifilament yarn is made from multiple filament fibres. Continuous filament fibre length requires little or no twisting to hold the multifilament yarn together.

Yarns are classified as simple, novelty and textured yarns.

a) Simple Yarns

Simple yarns are characterized by uniform size and regular surface. They can be broadly divided into single, ply, cord, and rope yarns.

i. **Single yarn:** This is the simplest type of yarn. It is commonly produced by twisting together staple or filament fibres.

ii. **Ply yarns:** These are produced by twisting two or more single yarns. Each strand of single yarn is referred to as a ply. Thus, four single yarns twisted together would form a four-ply yarn.

iii. **Cord yarns:** They are produced by twisting two or more ply yarns.

iv. **Rope yarns:** They are produced by twisting two or more cord yarns.

XII Std - Home Science EM_Unit 4.indd 104 20-04-2020 6.44.13 PM